What is the Best Geometry for Minimization of MMD?

Anonymous Author
Anonymous Institution

1 Introduction

We focus on MMD due to its interpretation as the worse-case integration error in the unit ball of a RKHS.

Notations: For a probability measure π, assume that $\int k(x,x)\,\mathrm{d}\pi(x) < \infty$. We define the associated kernel integral operator $\mathcal{T}_\pi : L_2(\pi) \to L_2(\pi)$ by $(\mathcal{T}_\pi f)(x) = \int k(x,y) f(y)\,\mathrm{d}\pi(y)$ for π-almost every x, where, as usual, elements of $L_2(\pi)$ are understood up to π-almost-everywhere equivalence.

A quote from Otto (Otto, 2001): The merit of the right gradient flow formulation of a dissipative evolution equation is that it separates energetics and kinetics: The energetics endow the state space with a functional, the kinetics endow the state space with a (Riemannian) geometry via the metric tensor

Assumption 1 (Kernel)**.** *Suppose the kernel is twice continuously differentiable; and the relevant kernel mean embedding is well-defined.*

2 Gradient Flows

Gradient flows characterize steepest-descent dynamics on a metric space. Let $(\mu_t)_{t\geq 0}$ be a curve of probability measures and let $\mathcal{F} : \mathcal{P}(\mathbb{R}^d) \to \mathbb{R}$ be an objective functional. The corresponding notion of steepest descent depends on the metric d imposed on $\mathcal{P}(\mathbb{R}^d)$. Consequently, a fixed objective $\mathcal{F}$ generally induces different gradient flows under different choices of d.

We consider the squared MMD objective $\mathcal{F}_{\mathrm{MMD}} = \frac{1}{2}\,\mathrm{MMD}^2(\cdot,\pi)$, motivated by its applications to numerical integration (Chen et al., 2026b) and generative modelling (Li et al., 2017; Arbel et al., 2018, 2019). We review the gradient flows associated with several standard geometries and discuss their asymptotic convergence in objective value, namely whether $\lim_{t\to\infty} \mathcal{F}_{\mathrm{MMD}}(\mu_t) = 0$. The present section focuses on population-level dynamics, while the next section develops finite-particle approximations. For later use, the first variation of $\mathcal{F}_{\mathrm{MMD}}$ at a measure μ is $\mathcal{F}'_{\mathrm{MMD}}[\mu](x) = \int k(x,y)\,\mathrm{d}(\mu-\pi)(y)$.

Wasserstein The 2-Wasserstein geometry is defined on $\mathcal{P}_2(\mathbb{R}^d)$, the space of probability measures with finite second moments. For $\mu,\nu \in \mathcal{P}_2(\mathbb{R}^d)$, the 2-Wasserstein distance is $d(\mu,\nu) = W_2(\mu,\nu) := \left(\inf_{\gamma\in\Pi(\mu,\nu)} \int \|x-y\|^2 \mathrm{d}\gamma(x,y)\right)^{\frac{1}{2}}$. Thus, $W_2^2(\mu,\nu)$ is the minimum quadratic cost of transporting μ to ν. Under this geometry, the Wasserstein gradient flow (W-GF) of $\mathcal{F}_{\mathrm{MMD}}$ is (Arbel et al., 2019; Korba et al., 2021)

$$\partial_t \mu_t = \nabla \cdot \left(\mu_t \nabla \mathcal{F}'_{\mathrm{MMD}}[\mu_t]\right). \tag{W-GF}$$

(W-GF) has been widely used for sampling (Chen et al., 2026b) and generative modelling (Galashov et al., 2025), with different kernel choices tailored to the application at hand. Despite its broad empirical success, establishing convergence of (W-GF) remains challenging in general. Several techniques have been proposed to improve its convergence behavior, including noise injection (Arbel et al., 2019; Suzuki et al., 2023; Chen et al., 2026a) and adaptive kernel lengthscales (Kang et al., 2026), but a general convergence theory is still lacking.

Fisher–Rao The Fisher–Rao geometry defines an alternative metric on the space of probability measures. Suppose that $\mu,\nu \in \mathcal{P}(\mathbb{R}^d)$ admit densities with respect to Lebesgue measure, and abuse the same symbols to denote these densities. Their Fisher–Rao distance is $d(\mu,\nu) = d_{\mathrm{FR}}(\mu,\nu) := 2\arccos\left(\int \sqrt{\mu(x)\nu(x)}\,\mathrm{d}x\right)$. The corresponding Fisher–Rao gradient flow (FR-GF) of $\mathcal{F}_{\mathrm{MMD}}$ is

$$\partial_t \mu_t = -\mu_t \left(\mathcal{F}'_{\mathrm{MMD}}[\mu_t] - \int \mathcal{F}'_{\mathrm{MMD}}[\mu_t]\mathrm{d}\mu_t\right). \tag{FR-GF}$$

Wasserstein–Fisher–Rao The W-GF in (W-GF) transports mass through space, whereas the FR-GF in (FR-GF) redistributes mass locally without spatial transport. The Wasserstein–Fisher–Rao gradient flow (WFR-GF) combines these two mechanisms. For $\alpha,\beta > 0$, the resulting dynamics are

$$\partial_t \mu_t = \alpha \nabla \cdot \left(\mu_t \nabla \mathcal{F}'_{\mathrm{MMD}}[\mu_t]\right)$$

$$-\beta\mu_t\left(\mathcal{F}'_{\mathrm{MMD}}[\mu_t]-\int\mathcal{F}'_{\mathrm{MMD}}[\mu_t]\mathrm{d}\mu_t\right). \tag{WFR-GF}$$

Wasserstein Interaction Force Transport Recent work has considered the geometry induced by the MMD distance, $d(\mu,\nu)=\mathrm{MMD}(\mu,\nu)$ (Zhu, 2024; Zhu and Mielke, 2024; Gladin et al., 2024). For a kernel k and a probability measure μ, let $\mathcal{T}_\mu : L^2(\mu)\to L^2(\mu)$ denote the associated kernel integral operator. The first variation of the MMD objective can be written as $\mathcal{F}'_{\mathrm{MMD}}=\int k(\cdot,y)\,\mathrm{d}(\mu-\pi)(y)=\mathcal{T}_\mu(1-\mathrm{d}\mu/\mathrm{d}\pi)$. The resulting Interaction-Force Transport (IFT) gradient flow is (Gladin et al., 2024)

$$\partial_t\mu_t=-\mu_t\mathcal{T}_{\mu_t}^{-1}\mathcal{F}'_{\mathrm{MMD}}[\mu_t]=-(\mu_t-\pi). \tag{IFT-GF}$$

This representation relies on the cancellation of $\mathcal{T}_{\mu_t}^{-1}$ with $\mathcal{T}_{\mu_t}$ in $\mathcal{F}'_{\mathrm{MMD}}$. The present work is restricted to $\mathcal{F}_{\mathrm{MMD}}$; extending this construction to more general objectives $\mathcal{F}$ is beyond its scope. (IFT-GF) redistributes mass without any spatial movement.

Combining W-GF and IFT-GF, analogously to (WFR-GF), yields, for $\alpha,\beta>0$,

$$\partial_t\mu_t=\alpha\nabla\cdot(\mu_t\nabla\mathcal{F}'_{\mathrm{MMD}}[\mu_t])-\beta(\mu_t-\pi). \tag{WIFT-GF}$$

Both IFT-GF and WIFT-GF enjoy nice convergence properties (Gladin et al., 2024); $\mathrm{MMD}^2(\mu_t,\pi)\le e^{-2\beta t}\,\mathrm{MMD}^2(\mu_0,\pi)$. Notably, this convergence holds without conditions on the target distribution π. The closest comparable result is Tian et al. (2026, Theorem 4.1), yet, it requires additional regularizations.

3 Particle Descent

In the last section, we have discussed the gradient flows on $\mathcal{P}(\mathbb{R}^d)$ with different geometries. Now, we are about to study their corresponding implementations with finite particles, i.e., particle descent scheme. Let $\mu_{N,t}=\sum_{i=1}^N w_t^{(i)}\delta_{x_t^{(i)}}$ be a weighted empirical measure at iteration t. We use bold symbols $\boldsymbol{X}_t=(x_t^{(1)},\ldots,x_t^{(N)})\in\mathbb{R}^{N\times d}$ and $\boldsymbol{w}_t=(w_t^{(1)},\ldots,w_t^{(N)})\in\mathbb{R}^N$. Denote $m_\pi=\int k(x,\cdot)\mathrm{d}\pi$ as the kernel mean embedding. Denote the kernel-mean residual and its vectorized counterpart as,

$$h_{\boldsymbol{X},\boldsymbol{w}}(x):=\sum_{j=1}^N w^{(j)}k(x^{(j)},x)-m_\pi(x)\in\mathbb{R},$$
$$h_{\boldsymbol{X},\boldsymbol{w}}(\boldsymbol{v}):=[h_{\boldsymbol{X},\boldsymbol{w}}(v^{(1)}),\ldots,h_{\boldsymbol{X},\boldsymbol{w}}(v^{(N)})]^\top\in\mathbb{R}^N.$$

Define the objective $\mathcal{E}(\boldsymbol{X},\boldsymbol{w})=\frac{1}{2}\mathrm{MMD}^2\left(\sum_{i=1}^N w^{(i)}\delta_{x^{(i)}},\pi\right)$. For simplicity, throughout this section we set $\alpha=\beta=1$.

Euclidean The most straightforward approach to optimizing the particle locations and weights is to apply gradient descent jointly to $\boldsymbol{X}$ and $\boldsymbol{w}$. Let $\tau_X,\tau_w>0$ be the step sizes for the particle locations and weights, respectively. We consider $\boldsymbol{X}_{t+1}=\boldsymbol{X}_t-\tau_X\nabla_{\boldsymbol{X}}\mathcal{E}(\boldsymbol{X}_t,\boldsymbol{w}_t)$ and $\boldsymbol{w}_{t+1}=\boldsymbol{w}_t-\tau_w\nabla_{\boldsymbol{w}}\mathcal{E}(\boldsymbol{X}_t,\boldsymbol{w}_t)$. For consistency with the update schemes introduced later, we adopt a Gauss–Seidel-type block gradient descent scheme, in which the particle locations are updated first and the weights are subsequently updated using the new locations $\boldsymbol{X}_{t+1}$. In particular, for each $i\in[N]$,

$$\begin{aligned}x_{t+1}^{(i)}&=x_t^{(i)}-\tau_X w_t^{(i)}\nabla h_{\boldsymbol{X}_t,\boldsymbol{w}_t}\left(x_t^{(i)}\right),\\ w_{t+1}^{(i)}&=w_t^{(i)}-\tau_w h_{\boldsymbol{X}_{t+1},\boldsymbol{w}_t}\left(x_{t+1}^{(i)}\right).\end{aligned} \tag{E-PD}$$

Note that the above descent scheme guarantees neither $w_i>0$ or $\sum_{i=1}^N w_i=1$.

Wasserstein Fisher-Rao Let $\tau_X,\tau_w>0$ be the location and weight step sizes. A finite-particle implementation of (WFR-GF) is: for each $i\in[N]$,

$$\begin{aligned}x_{t+1}^{(i)}&=x_t^{(i)}-\tau_X\nabla h_{\boldsymbol{X}_t,\boldsymbol{w}_t}\left(x_t^{(i)}\right),\\ w_{t+1}^{(i)}&=w_t^{(i)}-\tau_w h_{\boldsymbol{X}_{t+1},\boldsymbol{w}_t}\left(x_{t+1}^{(i)}\right).\end{aligned} \tag{WFR-PD}$$

When $\tau_w=0$, the weights remain fixed and hence the above dynamics reduce to the finite-particle implementation of MMD flow in Arbel et al. (2019). Another implementation of the Wasserstein Fisher–Rao gradient flow is based on fixed-point equations, called mean shift interacting particles (MSIP). Specifically, they directly solve the stationary conditions of (WFR-PD). The stationarity of the weight dynamics gives $h_{\boldsymbol{X},\boldsymbol{w}}(x^{(i)})=0$, which implies $\boldsymbol{w}=\boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X})=K(\boldsymbol{X},\boldsymbol{X})^{-1}m_\pi(\boldsymbol{X})$, i.e., the kernel quadrature weights associated with the current particle locations. Substituting these weights into the stationarity condition for the particle locations yields a fixed-point equation for $\boldsymbol{X}$, which MSIP solves iteratively.

[I am not too sure about MSIP. Maybe worth a meeting to discuss?]

Wasserstein IFT The locations are first updated in the same way as (E-PD) and (WFR-PD) above. However, the weight update scheme requires knowledge of the true density π, which is often intractable in practice. Following Gladin et al. (2024), the weights are updated instead via an implicit minimizing scheme. Let $\tau_X,\tau_w>0$ be the location and weight step sizes.

$$x_{t+1}^{(i)}=x_t^{(i)}-\tau_X\nabla h_{\boldsymbol{X}_t,\boldsymbol{w}_t}\left(x_t^{(i)}\right), \tag{WIFT-PD}$$

$$\boldsymbol{w}_{t+1}=\arg\min_{\boldsymbol{w}}\left\{\frac{1}{2}\mathrm{MMD}^2\left(\sum_{i=1}^N w^{(i)}\delta_{x_{t+1}^{(i)}},\pi\right)\right.$$

$$+\frac{1}{2\tau_w}\,\mathrm{MMD}^2\left(\sum_{i=1}^{N} w^{(i)}\delta_{x^{(i)}_{t+1}},\mu_{N,t}\right)\Bigg\}.$$

Interestingly, letting $K_{t+1} = K(\boldsymbol{X}_{t+1}, \boldsymbol{X}_{t+1}) \in \mathbb{R}^{N\times N}$ be the Gram matrix of the particles at iteration $t+1$, the weight update problem is equivalent to

$$\begin{aligned}\boldsymbol{w}_{t+1} &= \arg\min_{\boldsymbol{w}}\left\{\frac{1}{2}\boldsymbol{w}^\top K_{t+1}\boldsymbol{w} - m_\pi(\boldsymbol{X}_{t+1})^\top\boldsymbol{w}\right.\\ &\qquad\left. + \frac{1}{2\tau_w}\|\boldsymbol{w}-\boldsymbol{w}_t\|^2_{K_{t+1}}\right\}\\ &= \boldsymbol{w}_t - \frac{\tau_w}{1+\tau_w}K_{t+1}^{-1}h_{\boldsymbol{X}_{t+1},\boldsymbol{w}_t}(\boldsymbol{X}_{t+1})\\ &= \frac{1}{1+\tau_w}\boldsymbol{w}_t + \frac{\tau_w}{1+\tau_w}\boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1}).\end{aligned}$$

The last line holds by the definition of kernel quadrature: $m_\pi(\boldsymbol{X}_{t+1}) = K_{t+1}\boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})$.

3.1 Which update scheme is better?

Having introduced the update schemes (E-PD), (WFR-PD), and (WIFT-PD), we are now ready to compare their optimization behavior. Although all three schemes minimize the same objective, $\mathcal{E}(\boldsymbol{X},\boldsymbol{w}) = \frac{1}{2}\,\mathrm{MMD}^2\left(\sum_{i=1}^N w^{(i)}\delta_{x^{(i)}},\pi\right)$, they exhibit different optimization properties as a consequence of the different underlying geometries.

To this end, we first define weighted euclidean geometry and the relevant smoothness.

Definition 3.1 (Weighted Euclidean Geometry)**.** *Let $m \in \mathbb{R}$ and let $G \in \mathbb{R}^{m\times m}$ be a symmetric positive definite matrix. We equip the space $\mathbb{R}^m$ with the G-weighted Euclidean inner product $\langle u, v\rangle_G := u^\top G v$ and with the induced norm $\|u\|_G^2 := u^\top G u$. For a differentiable function $f : \mathbb{R}^m \to \mathbb{R}$, its gradient with respect to the G-weighted Euclidean geometry, denoted by $\mathrm{grad}_G f$, is defined as: for any $u \in \mathbb{R}^m$,*

$$\frac{\mathrm{d}}{\mathrm{d}t}f(x+tu)\bigg|_{t=0} = \langle \mathrm{grad}_G f(x), u\rangle_G .$$

Equivalently, $\mathrm{grad}_G f(x) = G^{-1}\nabla f(x)$ where ∇f denotes the standard Euclidean gradient.

Definition 3.2 (Smoothness)**.** *A differentiable function $f : \mathbb{R}^m \to \mathbb{R}$ is said to be L-smooth with respect to the G-weighted Euclidean geometry if $\|\mathrm{grad}_G f(x) - \mathrm{grad}_G f(y)\|_G \le L\|x-y\|_G$ for all $x, y \in \mathbb{R}^m$.*

Proposition 3.3 (Smoothness of squared MMD objective under various geometries)**.** *Let $\boldsymbol{w} \in \mathbb{R}^N$ with $w^{(i)} > 0$ for every i and $\sum_{i=1}^N w^{(i)} = 1$. Let $\boldsymbol{X} \in \mathbb{R}^{N\times d}$ and suppose that the associated Gram matrix $K(\boldsymbol{X},\boldsymbol{X}) \in \mathbb{R}^{N\times N}$ is positive definite. Suppose that $\sup_{x,y\in\mathbb{R}^d}\|\nabla^2_{11}k(x,y)\|_{\mathrm{op}} = C_1 < \infty$ and $\sup_{x,y\in\mathbb{R}^d}\|\nabla^2_{12}k(x,y)\|_{\mathrm{op}} = C_2 < \infty$. Then,*

1. *$\mathcal{E}(\cdot,\boldsymbol{w})$ is $\max_{1\le i\le N} w^{(i)}(2C_1+C_2)$-smooth under the standard Euclidean geometry; and is $(2C_1+C_2)$-smooth under the $\mathrm{Diag}(\boldsymbol{w})$-weighted Euclidean geometry.*
2. *$\mathcal{E}(\boldsymbol{X},\cdot)$ is $\lambda_{\max}(K(\boldsymbol{X},\boldsymbol{X}))$-smooth under the standard Euclidean geometry; and is 1-smooth under the $K(\boldsymbol{X},\boldsymbol{X})$-weighted Euclidean geometry.*

The above proposition shows that the smoothness of the squared MMD objective depends strongly on the underlying geometry. In particular, an appropriate choice of geometry can remove the dependence of smoothness L on quantities such as the particle weights $w^{(i)}$ or the spectrum of the Gram matrix. One important implication of smoothness is that it determines the admissible scale of a gradient step. For an L-smooth objective, the descent lemma shows that a step size no larger than $2/L$ is sufficient for the objective to be non-increasing. We next analyze the particle descent schemes introduced above, identify the geometry underlying each scheme, and compare their corresponding smoothness properties and admissible step sizes.

Proposition 3.4 (Sufficient step sizes for monotone descent)**.** *Suppose that the assumptions of Proposition 3.3 hold with $\boldsymbol{w} = \boldsymbol{w}_t$ and $\boldsymbol{X} = \boldsymbol{X}_{t+1}$. Then each particle descent scheme satisfies $\mathcal{E}(\boldsymbol{X}_{t+1},\boldsymbol{w}_{t+1}) \le \mathcal{E}(\boldsymbol{X}_t,\boldsymbol{w}_t)$ under the following sufficient conditions:*

1. *For E-PD, the sufficient location condition is $\tau_X \le 2/[\max_{1\le i\le N} w_t^{(i)}(2C_1+C_2)]$. The sufficient weight condition is $\tau_w \le 2/\lambda_{\max}(K_{t+1})$.*
2. *For WFR-PD, the sufficient location condition is $\tau_X \le 2/(2C_1+C_2)$. The sufficient weight condition is $\tau_w \le 2/\lambda_{\max}(K_{t+1})$.*
3. *For MSIP, XXX.*
4. *For WIFT-PD, the sufficient location condition is $\tau_X \le 2/(2C_1+C_2)$. The sufficient weight condition is $0 < \tau_w$.*

The different conditions on the step sizes in Proposition 3.4 follow directly from the underlying geometries and the correspoinding smoothness proved in Proposition 3.3. The location update of (E-PD) uses the standard Euclidean geometry, whereas those of (WFR-PD) and (WIFT-PD) use the $\mathrm{Diag}(\boldsymbol{w})$-weighted Euclidean geometry. The weight updates of (E-PD) and (WFR-PD) use the standard Euclidean geometry, whereas the implicit weight update of (WIFT-PD) is proximal in the $K(\boldsymbol{X},\boldsymbol{X})$-weighted Euclidean geometry. Separate step sizes conditions are therefore almost immediate because the location and weight blocks generally have different smoothness constants established in Proposition 3.3.

Remark 3.5 (WIFT-PD as a favorable geometry)**.** *The main message of this paper is that WIFT-PD provides a particularly favorable geometry for finite-particle MMD minimization. While its updates can be interpreted as preconditioned gradient steps and all the step-size conditions follow naturally, our contribution is to provide, to the best of our knowledge, the first unified finite-particle analysis of the different geometries and their associated preconditioners. This perspective makes explicit how the choice of geometry affects the particle dynamics, smoothness properties, admissible step sizes, and convergence behavior.*

[The following is Work in Progress.]

4 Convergence

Theorem 4.1 (Convergence)**.** *Let the iterates* $\mu_{N,t} = \sum_{i=1}^{N} w_t^{(i)} \delta_{x_t^{(i)}}$*, with* $\boldsymbol{X}_t = (x_t^{(1)}, \dots, x_t^{(N)})$ *and* $\boldsymbol{w}_t = (w_t^{(1)}, \dots, w_t^{(N)})$*, be generated by WIFT-PD with step sizes* $\tau_X, \tau_w > 0$*. The initial weights are chosen as* $\boldsymbol{w}_0 = \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_0)$*. If* $\tau_X \leq 2(2C_1 + C_2)^{-1}$*, then for every* $t \geq 0$*,*

$$\mathcal{E}\left(\boldsymbol{X}_t, \boldsymbol{w}_t\right) \leq \frac{1}{(1+\tau_w)^{2t}} \mathcal{E}(\boldsymbol{X}_0, \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_0)) + \sum_{s=1}^{t} \left(\frac{1}{(1+\tau_w)^{2(t-s)}} - \frac{1}{(1+\tau_w)^{2(t-s+1)}} \right) \mathcal{E}(\boldsymbol{X}_s, \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_s)).$$

See the proof of Theorem 4.1. The above theorem shows that the squared MMD error at iteration t is controlled by a convex combination of the kernel-quadrature residuals associated with the particle locations visited from initialization up to iteration $t-1$.

Importantly, the contribution of earlier particle configurations decays geometrically: the influence of an iterate that is r steps in the past is proportional to $(1+\tau_w)^{-2r}$. Thus, WIFT progressively "forgets" earlier particle configurations and places increasingly more weight on the kernel-quadrature quality of the recent iterates. Another notable feature of this geometric forgetting is that its rate is independent of the condition number of the kernel Gram matrix $\lambda_{\max}(k(\boldsymbol{X}_t, \boldsymbol{X}_t))$. This contrasts with the Euclidean weight updates in Euc-PD and WFR-PD, whose convergence behavior generally depends on the spectrum of the Gram matrix.

Under the same step-size condition, WIFT-PD also satisfies the monotone decrease $\mathrm{MMD}(\mu_{N,t+1}, \pi) \leq \mathrm{MMD}(\mu_{N,t}, \pi)$; see Proposition A.1 in the appendix.

5 Experiments

We compare Euc-PD, WFR, WIFT, and MSIP for the two-dimensional target $\pi = \mathcal{N}(\mathbf{0}, I_2)$. Every run uses $N = 20$ particles, seed 42, RBF length scale 1, and 1000 iterations. The close and far initial distributions are $\mathcal{N}((1,1), I_2)$ and $\mathcal{N}((9,9), I_2)$, respectively. Euc-PD, WFR, and WIFT use the linear-plus-RBF kernel with unit linear coefficient, whereas the upstream MSIP implementation uses the RBF kernel. We use $\alpha = 1$ and $\beta = 0.5$ and report unsmoothed trajectories.

Figure 1 compares $\tau \in \{0.01, 0.1, 1\}$. The MSIP curve for $\tau = 1$ is omitted because the upstream update encounters a singular Gram matrix in both initialization settings. Appendix B provides the corresponding particle trajectories.

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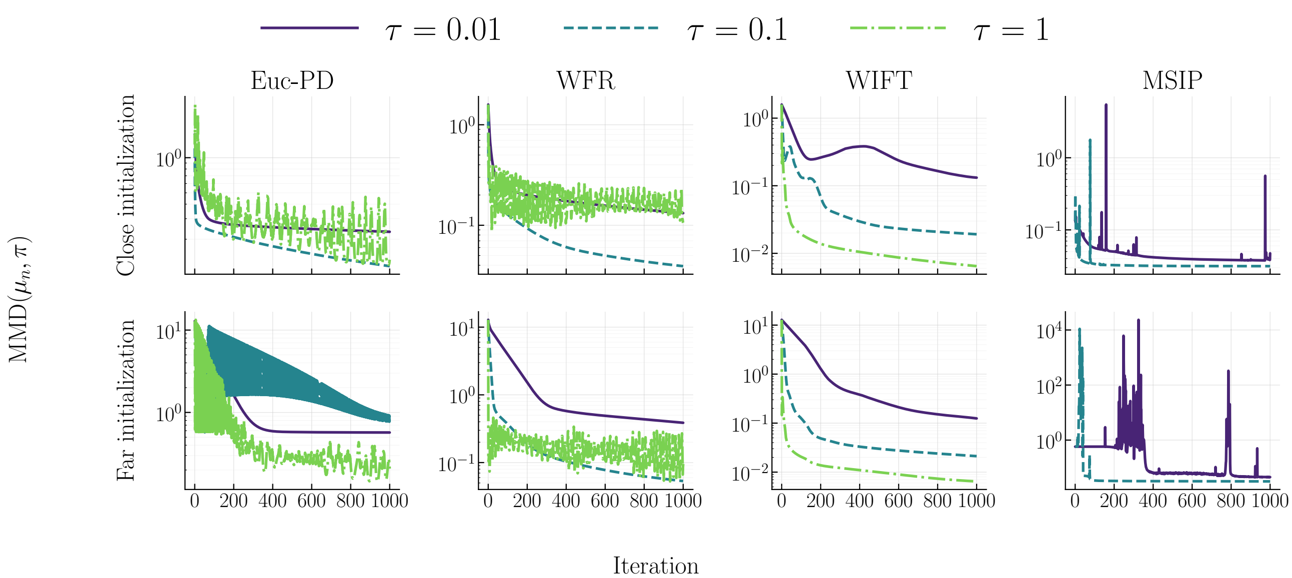

Figure 1: **Close and far Gaussian initialization.** Unsmoothed MMD trajectories for Euc-PD, WFR, WIFT, and MSIP with target $\pi = \mathcal{N}(\mathbf{0}, I_2)$. The rows use initial distributions $\mathcal{N}((1,1), I_2)$ and $\mathcal{N}((9,9), I_2)$, respectively.

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A Proofs

A.1 Proofs in Section 2

Proof of Proposition 3.3. Recall that

$$E(\boldsymbol{X}, \boldsymbol{w}) = \frac{1}{2} \sum_{i,j=1}^{N} w^{(i)} w^{(j)} k(x^{(i)}, x^{(j)}) - \sum_{i=1}^{N} w^{(i)} m_\pi(x^{(i)}) + \text{const.}$$

For any $\boldsymbol{u} = (u^{(1)}, \ldots, u^{(N)}) \in (\mathbb{R}^d)^N$, direct differentiation gives

$$\begin{aligned} \nabla^2_{\boldsymbol{X}} E(\boldsymbol{X}, \boldsymbol{w})[\boldsymbol{u}, \boldsymbol{u}] = &\sum_{i,j=1}^{N} w^{(i)} w^{(j)} (u^{(i)})^\top \nabla^2_{11} k(x^{(i)}, x^{(j)}) u^{(i)} \\ &+ \sum_{i,j=1}^{N} w^{(i)} w^{(j)} (u^{(i)})^\top \nabla^2_{12} k(x^{(i)}, x^{(j)}) u^{(j)} - \sum_{i=1}^{N} w^{(i)} (u^{(i)})^\top \nabla^2 m_\pi(x^{(i)}) u^{(i)}. \end{aligned}$$

Since $m_\pi(x) = \int k(x, y) \,\mathrm{d}\pi(y)$ and π is a probability measure, we have

$$\|\nabla^2 m_\pi(x)\|_{\text{op}} = \left\| \int \nabla^2_{11} k(x, y) \,\mathrm{d}\pi(y) \right\|_{\text{op}} \le \sup_{x,y \in \mathbb{R}^d} \|\nabla^2_{11} k(x, y)\|_{\text{op}} = C_1.$$

Using $\sum_{j=1}^{N} w^{(j)} = 1$, it follows that

$$\left| \sum_{i,j=1}^{N} w^{(i)} w^{(j)} (u^{(i)})^\top \nabla^2_{11} k(x^{(i)}, x^{(j)}) u^{(i)} - \sum_{i=1}^{N} w^{(i)} (u^{(i)})^\top \nabla^2 m_\pi(x^{(i)}) u^{(i)} \right| \le 2C_1 \sum_{i=1}^{N} w^{(i)} \|u^{(i)}\|^2.$$

For the mixed term,

$$\begin{aligned} \left| \sum_{i,j=1}^{N} w^{(i)} w^{(j)} (u^{(i)})^\top \nabla^2_{12} k(x^{(i)}, x^{(j)}) u^{(j)} \right| &\le \sup_{x,y \in \mathbb{R}^d} \|\nabla^2_{12} k(x, y)\|_{\text{op}} \left(\sum_{i=1}^{N} w^{(i)} \|u^{(i)}\| \right)^2 \\ &= C_2 \left(\sum_{i=1}^{N} w^{(i)} \|u^{(i)}\| \right)^2. \end{aligned}$$

By Cauchy–Schwarz and $\sum_{i=1}^{N} w^{(i)} = 1$, we have $\left(\sum_{i=1}^{N} w^{(i)} \|u^{(i)}\| \right)^2 \le \sum_{i=1}^{N} w^{(i)} \|u^{(i)}\|^2$. Therefore,

$$\left| \nabla^2_{\boldsymbol{X}} E(\boldsymbol{X}, \boldsymbol{w})[\boldsymbol{u}, \boldsymbol{u}] \right| \le (2C_1 + C_2) \sum_{i=1}^{N} w^{(i)} \|u^{(i)}\|^2 = (2C_1 + C_2) \|\boldsymbol{u}\|^2_{\boldsymbol{w}}.$$

The last step holds by the definition of $\boldsymbol{w}$-weighted Euclidean geometry in Definition 3.1. This proves smoothness under the weighted Euclidean geometry.

For the standard Euclidean geometry, since $\sum_{i=1}^{N} w^{(i)} \|u^{(i)}\|^2 \le \left(\max_{1 \le i \le N} w^{(i)} \right) \sum_{i=1}^{N} \|u^{(i)}\|^2 = \left(\max_{1 \le i \le N} w^{(i)} \right) \|\boldsymbol{u}\|^2_2$. This proves the first claim.

For fixed $\boldsymbol{X}$, the weight gradient is $\nabla_{\boldsymbol{w}} \mathcal{E}(\boldsymbol{X}, \boldsymbol{w}) = K(\boldsymbol{X}, \boldsymbol{X}) \boldsymbol{w} - m_\pi(\boldsymbol{X})$. Hence, for any $\boldsymbol{w}, \boldsymbol{v} \in \mathbb{R}^N$, we have $\|\nabla_{\boldsymbol{w}} \mathcal{E}(\boldsymbol{X}, \boldsymbol{w}) - \nabla_{\boldsymbol{w}} \mathcal{E}(\boldsymbol{X}, \boldsymbol{v})\|_2 \le \lambda_{\max}(K(\boldsymbol{X}, \boldsymbol{X})) \|\boldsymbol{w} - \boldsymbol{v}\|_2$. Under the $K(\boldsymbol{X}, \boldsymbol{X})$-weighted Euclidean geometry, the corresponding gradient is $\operatorname{grad}_K \mathcal{E}(\boldsymbol{X}, \boldsymbol{w}) = \boldsymbol{w} - K(\boldsymbol{X}, \boldsymbol{X})^{-1} m_\pi(\boldsymbol{X})$, and therefore $\|\operatorname{grad}_K \mathcal{E}(\boldsymbol{X}, \boldsymbol{w}) - \operatorname{grad}_K \mathcal{E}(\boldsymbol{X}, \boldsymbol{v})\|_K = \|\boldsymbol{w} - \boldsymbol{v}\|_K$. This proves the second claim. $\square$

Proof of Proposition 3.4. For an L-smooth function f under a fixed weighted Euclidean geometry, the descent lemma applied to $z^+ = z - \gamma \operatorname{grad}_G f(z)$ gives

$$f(z^+) \le f(z) - \gamma \left(1 - \frac{L\gamma}{2} \right) \|\operatorname{grad}_G f(z)\|^2_G.$$

Thus, the corresponding block objective is non-increasing whenever $\gamma \le 2/L$.

For E-PD, the location substep is standard Euclidean gradient descent with effective step size τ_X. By Proposition 3.3, its smoothness constant is $\max_i w_t^{(i)}(2C_1 + C_2)$. With $\boldsymbol{X}_{t+1}$ fixed, the weight substep is standard Euclidean gradient descent with effective step size τ_w and smoothness constant $\lambda_{\max}(K_{t+1})$. The stated bounds therefore imply

$$\mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}_{t+1}) \le \mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}_t) \le \mathcal{E}(\boldsymbol{X}_t, \boldsymbol{w}_t).$$

For WFR-PD, the location substep is gradient descent under the $\mathrm{Diag}(\boldsymbol{w}_t)$-weighted Euclidean geometry with step size τ_X and smoothness constant $2C_1 + C_2$. Its weight substep is standard Euclidean gradient descent with step size τ_w and smoothness constant $\lambda_{\max}(K_{t+1})$. The same descent argument gives the claimed monotonicity.

The WIFT-PD location substep is identical to that of WFR-PD and is therefore non-increasing under the stated location condition. For the implicit weight substep, its defining minimization problem and the admissible choice $\boldsymbol{w} = \boldsymbol{w}_t$ give

$$\mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}_{t+1}) + \frac{1}{2\tau_w}\|\boldsymbol{w}_{t+1} - \boldsymbol{w}_t\|^2_{K_{t+1}} \le \mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}_t).$$

The squared norm is nonnegative for every $\tau_w > 0$, so the weight substep is non-increasing without an upper bound on τ_w. Combining the two substeps concludes the proof. □

A.2 Proofs in Section 3

Proposition A.1 (Monotone decrease of WIFT-PD)**.** *Let the iterates $\mu_{N,t} = \sum_{i=1}^N w_t^{(i)} \delta_{x_t^{(i)}}$, with $\boldsymbol{X}_t = (x_t^{(1)}, \ldots, x_t^{(N)})$ and $\boldsymbol{w}_t = (w_t^{(1)}, \ldots, w_t^{(N)})$, be generated by WIFT-PD with step sizes $\tau_X, \tau_w > 0$. Suppose that the assumptions of Proposition 3.3 hold with $\boldsymbol{w} = \boldsymbol{w}_t$ and $\boldsymbol{X} = \boldsymbol{X}_{t+1}$ at every iteration, and that $\tau_X \le 2(2C_1 + C_2)^{-1}$. Then, for every $t \ge 0$,*

$$\mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}_{t+1}) \le \mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}_t) \le \mathcal{E}(\boldsymbol{X}_t, \boldsymbol{w}_t),$$

and hence $\mathrm{MMD}(\mu_{N,t+1}, \pi) \le \mathrm{MMD}(\mu_{N,t}, \pi)$.

Proof of Proposition A.1. For fixed $\boldsymbol{X}$ and $\boldsymbol{w}$, let $g^{(i)}(\boldsymbol{X}, \boldsymbol{w}) = \sum_{j=1}^N w^{(j)} \nabla_2 k(x^{(j)}, x^{(i)}) - \nabla m_\pi(x^{(i)})$. Define the squared MMD energy by

$$\mathcal{E}(\boldsymbol{X}, \boldsymbol{w}) = \frac{1}{2}\,\mathrm{MMD}^2\left(\sum_{i=1}^N w^{(i)} \delta_{x^{(i)}}, \pi\right), \tag{1}$$

Differentiating the squared MMD energy with respect to $x^{(i)}$ and using symmetry of the kernel gives $\nabla_{x^{(i)}} \mathcal{E}(\boldsymbol{X}, \boldsymbol{w}) = w^{(i)} g^{(i)}(\boldsymbol{X}, \boldsymbol{w})$. Hence, the position update of (WIFT-PD) can be written as $x_{t+1}^{(i)} = x_t^{(i)} - \frac{\tau_X}{w_t^{(i)}} \nabla_{x^{(i)}} \mathcal{E}(\boldsymbol{X}_t, \boldsymbol{w}_t)$. By the $(2C_1 + C_2)$-smoothness of $\mathcal{E}(\cdot, \boldsymbol{w}_t)$ with respect to the $\mathrm{Diag}(\boldsymbol{w}_t)$-weighted Euclidean geometry established in Proposition 3.3, the descent lemma gives

$$\begin{aligned}
\mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}_t) &\le \mathcal{E}(\boldsymbol{X}_t, \boldsymbol{w}_t) - \tau_X \left(1 - \frac{(2C_1 + C_2)\tau_X}{2}\right) \left\|\mathrm{grad}_{\mathrm{Diag}(\boldsymbol{w}_t)}\, \mathcal{E}(\boldsymbol{X}_t, \boldsymbol{w}_t)\right\|^2_{\mathrm{Diag}(\boldsymbol{w}_t)} \\
&= \mathcal{E}(\boldsymbol{X}_t, \boldsymbol{w}_t) - \tau_X \left(1 - \frac{(2C_1 + C_2)\tau_X}{2}\right) \sum_{i=1}^N w_t^{(i)} \left\|g^{(i)}(\boldsymbol{X}_t, \boldsymbol{w}_t)\right\|^2 .
\end{aligned} \tag{2}$$

Since $\tau_X \le 2(2C_1 + C_2)^{-1}$, this proves that $\mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}_t) \le \mathcal{E}(\boldsymbol{X}_t, \boldsymbol{w}_t)$ so the location update in (WIFT-PD) decreases the MMD value.

We next consider the weight update. For fixed $\boldsymbol{X}$, the dependence of the MMD energy on $\boldsymbol{w}$ is quadratic: $\mathcal{E}(\boldsymbol{X}, \boldsymbol{w}) = \frac{1}{2}\boldsymbol{w}^\top K(\boldsymbol{X})\boldsymbol{w} - m_\pi(\boldsymbol{X})^\top \boldsymbol{w} + \text{const}$. Since $K(\boldsymbol{X})$ is positive definite, the unique minimizer is $\boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}) = K(\boldsymbol{X})^{-1} m_\pi(\boldsymbol{X})$. Completing the square gives

$$\mathcal{E}(\boldsymbol{X}, \boldsymbol{w}) = \mathcal{E}\left(\boldsymbol{X}, \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X})\right) + \frac{1}{2}\left(\boldsymbol{w} - \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X})\right)^\top K(\boldsymbol{X})\left(\boldsymbol{w} - \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X})\right). \tag{3}$$

Hence for fixed particle locations $\boldsymbol{X}_{t+1}$, the KQ decomposition gives

$$\begin{aligned}&\mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}_{t+1}) - \mathcal{E}\left(\boldsymbol{X}_{t+1}, \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})\right) \\ &\quad = \frac{1}{2}\left(\boldsymbol{w}_{t+1} - \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})\right)^\top K_{t+1}\left(\boldsymbol{w}_{t+1} - \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})\right). \end{aligned} \tag{4}$$

Also, by the weight update,

$$\boldsymbol{w}_{t+1} - \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1}) = \frac{1}{1+\tau_w}\left(\boldsymbol{w}_t - \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})\right). \tag{5}$$

Substituting Eq. (5) into Eq. (4) yields

$$\begin{aligned}&\mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}_{t+1}) - \mathcal{E}\left(\boldsymbol{X}_{t+1}, \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})\right) \\ &\quad = \frac{1}{2(1+\tau_w)^2}\left(\boldsymbol{w}_t - \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})\right)^\top K_{t+1}\left(\boldsymbol{w}_t - \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})\right) \\ &\quad = \frac{1}{(1+\tau_w)^2}\left[\mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}_t) - \mathcal{E}\left(\boldsymbol{X}_{t+1}, \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})\right)\right]. \end{aligned} \tag{6}$$

Since $(1+\tau_w)^{-2} \leq 1$ and $\boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})$ minimizes $\boldsymbol{w} \mapsto \mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w})$, it follows that $\mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}_{t+1}) \leq \mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}_t)$. Combining the two sub-steps yields

$$\mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}_{t+1}) \leq \mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}_t) \leq \mathcal{E}(\boldsymbol{X}_t, \boldsymbol{w}_t). \tag{7}$$

Hence, $\mathcal{E}(\boldsymbol{X}_t, \boldsymbol{w}_t)$ is nonincreasing, and the proof is concluded. □

Lemma A.2 (Uniform control of the kernel quadrature residual)**.** *Suppose that the assumptions of Proposition A.1 hold and that the initial weights are chosen as $\boldsymbol{w}_0 = \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_0)$. Then, for every $t \geq 0$,* $\mathrm{MMD}\left(\sum_{i=1}^N w_{\mathrm{KQ}}^{(i)}(\boldsymbol{X}_t)\delta_{x_t^{(i)}}, \pi\right) \leq \mathrm{MMD}\left(\sum_{i=1}^N w_{\mathrm{KQ}}^{(i)}(\boldsymbol{X}_0)\delta_{x_0^{(i)}}, \pi\right)$.

Proof of Lemma A.2. Recall the definition that $\mathcal{E}(\boldsymbol{X}, \boldsymbol{w}) = \frac{1}{2}\mathrm{MMD}^2\left(\sum_{i=1}^N w^{(i)}\delta_{x^{(i)}}, \pi\right)$. For any fixed particle locations $\boldsymbol{X}_t$, the kernel quadrature weights $\boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_t)$ minimize the MMD energy over the weights. Hence, $\mathcal{E}\left(\boldsymbol{X}_t, \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_t)\right) \leq \mathcal{E}\left(\boldsymbol{X}_t, \boldsymbol{w}_t\right)$. On the other hand, by the monotonicity of the WIFT iteration established in Proposition A.1, we have $\mathcal{E}\left(\boldsymbol{X}_t, \boldsymbol{w}_t\right) \leq \mathcal{E}\left(\boldsymbol{X}_0, \boldsymbol{w}_0\right)$. Since $\boldsymbol{w}_0 = \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_0)$, we have $\mathcal{E}(\boldsymbol{X}_0, \boldsymbol{w}_0) = \mathcal{E}(\boldsymbol{X}_0, \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_0))$. Combining the previous inequalities gives

$$\mathcal{E}\left(\boldsymbol{X}_t, \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_t)\right) \leq \mathcal{E}\left(\boldsymbol{X}_t, \boldsymbol{w}_t\right) \leq \mathcal{E}\left(\boldsymbol{X}_0, \boldsymbol{w}_0\right) = \mathcal{E}\left(\boldsymbol{X}_0, \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_0)\right). \tag{8}$$

The proof is thus concluded. □

Theorem A.3 (Convergence; verbatim of Theorem 4.1)**.** *Let the iterates $\mu_{N,t} = \sum_{i=1}^N w_t^{(i)}\delta_{x_t^{(i)}}$, with $\boldsymbol{X}_t = (x_t^{(1)}, \ldots, x_t^{(N)})$ and $\boldsymbol{w}_t = (w_t^{(1)}, \ldots, w_t^{(N)})$, be generated by WIFT-PD with step sizes $\tau_X, \tau_w > 0$. The initial weights are chosen as $\boldsymbol{w}_0 = \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_0)$. If $\tau_X \leq 2(2C_1 + C_2)^{-1}$, then for every $t \geq 0$,*

$$\mathcal{E}\left(\boldsymbol{X}_t, \boldsymbol{w}_t\right) \leq \frac{1}{(1+\tau_w)^{2t}}\mathcal{E}(\boldsymbol{X}_0, \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_0)) + \sum_{s=1}^t\left(\frac{1}{(1+\tau_w)^{2(t-s)}} - \frac{1}{(1+\tau_w)^{2(t-s+1)}}\right)\mathcal{E}(\boldsymbol{X}_s, \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_s)).$$

Proof of Theorem A.3. From the exact contraction of the weight update in Eq. (6), together with descent of the Wasserstein position update, we have

$$\mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}_{t+1}) \leq \frac{1}{(1+\tau_w)^2}\mathcal{E}(\boldsymbol{X}_t, \boldsymbol{w}_t) + \left(1 - \frac{1}{(1+\tau_w)^2}\right)\mathcal{E}\left(\boldsymbol{X}_{t+1}, \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})\right). \tag{9}$$

Applying the same iteration above recursively, then we obtain,

$$\begin{aligned}\mathcal{E}\left(\boldsymbol{X}_{t+1}, \boldsymbol{w}_{t+1}\right) &\leq \frac{1}{(1+\tau_w)^{2t}}\mathcal{E}(\boldsymbol{X}_0, \boldsymbol{w}_0) + \left(1 - \frac{1}{(1+\tau_w)^2}\right)\sum_{s=1}^t \frac{\mathcal{E}(\boldsymbol{X}_s, \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_s))}{(1+\tau_w)^{2(t-s)}}. \\ &= \frac{1}{(1+\tau_w)^{2t}}\mathcal{E}(\boldsymbol{X}_0, \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_0)) + \sum_{s=1}^t\left(\frac{1}{(1+\tau_w)^{2(t-s)}} - \frac{1}{(1+\tau_w)^{2(t-s+1)}}\right)\mathcal{E}(\boldsymbol{X}_s, \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_s)).\end{aligned}$$

The proof is concluded. □

B Close- and Far-Initialization Experiments

We compare two Gaussian initialization settings in two dimensions: close initialization from $\mathcal{N}((1,1), I_2)$ and far initialization from $\mathcal{N}((9,9), I_2)$. The target is $\pi = \mathcal{N}(\mathbf{0}, I_2)$, and every run uses $N = 20$ particles, seed 42, RBF length scale 1, and 1000 iterations. All methods use the same seed-controlled initial cloud in each setting, and the far cloud is the close cloud translated by $(8, 8)$. Euc-PD, WFR, and WIFT use the linear-plus-RBF kernel with unit linear coefficient, whereas the upstream MSIP implementation uses the RBF kernel. We use $\alpha = 1$ and $\beta = 0.5$ and retain the unsmoothed trajectories. Figure 1 reports the sweep over $\tau \in \{0.01, 0.1, 1\}$. The MSIP curve for $\tau = 1$ is omitted because the upstream update encounters a singular Gram matrix in both settings.

Figure 2 adapts the trajectory visualization of Gladin et al. (2024, Figures 4–5) to these two-dimensional experiments. Both axes are spatial coordinates, the blue contours represent the target density, and iteration progresses along each dotted path. Each panel shows all 20 particles: orange crosses mark step 0, and green circles mark step 1000. To keep the paths readable, we use $\tau = 0.01$ for Euc-PD and MSIP, $\tau = 0.1$ for WFR, and $\tau = 1$ for WIFT. Each row uses a common spatial window spanning the target and the corresponding Gaussian initial cloud; path segments outside that window are clipped.

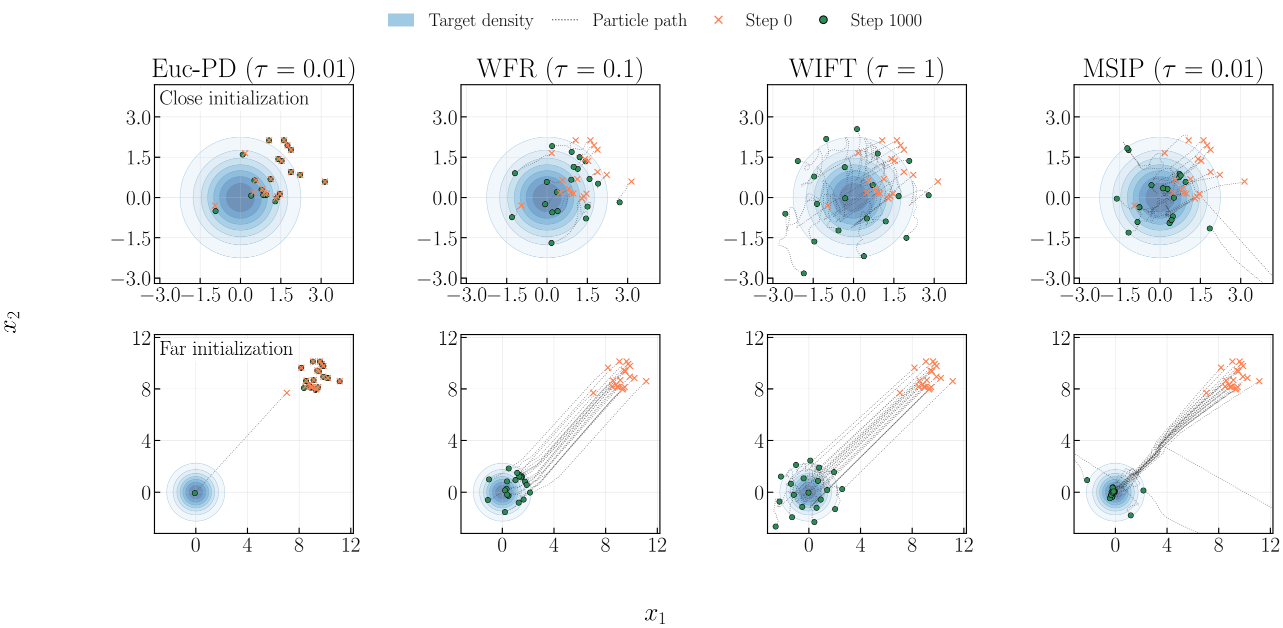

Figure 2: **Particle trajectories under close and far initialization.** Paths of all 20 particles from step 0 to step 1000 for initial distributions $\mathcal{N}((1,1), I_2)$ and $\mathcal{N}((9,9), I_2)$, respectively.